FINITE BIRATIONAL MORPHISMS TOWARD A NORMAL SPACE ARE ISOMORPHISMS OVER THE NORMAL LOCUS

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Editorial note. The source says 'and its proof' for the scheme case; the scheme lemma is cross-chapter and its statement (not proof) is used, declared external.

Allowed standard prerequisites (editorial).

- Birational induced morphism of normalizations. lemma-birational-induced-morphism-normalizations (carded L26753). The induced morphism of normalizations.
- Normalization of a normal space. lemma-normalization-normal (same chapter context; the and-its-proof pointer is NOT carded, declared external). The normalization of a normal space is itself.

Source reference. Original lemma 0B4D: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 5309–5309. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0AB1: focused-seeds/geometry/S9-decent-finite-birational-over-normal/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0357: focused-seeds/geometry/S9-decent-finite-birational-over-normal/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume*

(1) *X and Y are decent and have finitely many irreducible components,*
(2) *f is integral and birational,*
(3) *Y is normal, and*
(4) *X is reduced.*

Then f is an isomorphism.

Proof. Let $V \to Y$ be an étale morphism with V affine. It suffices to show that $U = X \times_Y V \to V$ is an isomorphism. By Lemma 0B4D and its proof we see that U and V are decent and have finitely many irreducible components and that $U \to V$

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is birational. By Properties, Lemma 0357 V is a finite disjoint union of integral schemes. Thus we may assume V is integral. As f is birational, we see that U is irreducible and reduced, i.e., integral (note that U is a scheme as f is integral, hence representable). Thus we may assume that X and Y are integral schemes and the result follows from the case of schemes, see Morphisms, Lemma 0AB1. $\square$